

CH9.1 指數與指數函數

$$1. a < b < c \quad 2.3 \quad 3.3 + 2\sqrt{2} \quad 4.\frac{3}{2} \quad 5.2$$

$$6.6 \quad 7. (1) -2 < x < 2 \quad (2) x < -1 \text{ 或 } x > 0 \quad 8. -5, -9 \quad 9. 5, 0 \quad 10. (-2, 3)$$

解析

$$1. \begin{cases} a = 2^{\frac{1}{4}} = (2^5)^{\frac{1}{20}} = 32^{\frac{1}{20}} \\ b = 3^{\frac{1}{5}} = (3^4)^{\frac{1}{20}} = 81^{\frac{1}{20}} \end{cases} \Rightarrow a < b$$

$$\begin{cases} b = 3^{\frac{1}{5}} = (3^6)^{\frac{1}{30}} = 729^{\frac{1}{30}} \\ c = 5^{\frac{1}{6}} = (5^5)^{\frac{1}{30}} = 3125^{\frac{1}{30}} \end{cases} \Rightarrow b < c$$

得 $a < b < c$

$$2. \begin{cases} 4^x = 12^z, 4 = 12^{\frac{z}{x}}, 4^3 = 12^{\frac{3z}{x}} \\ 27^y = 12^z, 27 = 12^{\frac{z}{y}}, 3^3 = 12^{\frac{z}{y}} \end{cases}$$

$$\text{相乘 } 4^3 \cdot 3^3 = 12^{\frac{3z}{x}} \cdot 12^{\frac{z}{y}}$$

$$12^3 = 12^{\frac{3z}{x} + \frac{z}{y}}, \frac{3z}{x} + \frac{z}{y} = 3$$

$$\frac{3}{x} + \frac{1}{y} = \frac{3}{z}, a = \frac{3}{2}$$

$$3. (1) 2^x + 2^y \geq 2\sqrt{2^x \cdot 2^y} = 2\sqrt{2^{x+y}} = 2\sqrt{2}$$

$$(2) \because x \geq 0, y \geq 0$$

$$\therefore 2^x \geq 1, 2^y \geq 1 \text{ 得 } (2^x - 1)(2^y - 1) \geq 0$$

$$\Rightarrow 2^x \cdot 2^y - 2^x - 2^y + 1 \geq 0 \Rightarrow 2^{x+y} + 1 \geq 2^x + 2^y$$

$$\text{即 } 2^x + 2^y \leq 2 + 1 = 3$$

$$\text{由(1)、(2)得 } 2\sqrt{2} \leq 2^x + 2^y \leq 3$$

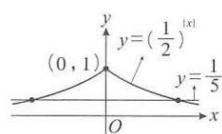
$$\therefore M + m = 3 + 2\sqrt{2}$$

$$4. 4^x - 5 \cdot 2^{x+1} + 2\sqrt{2} = 0 \Rightarrow (2^x)^2 - 10 \cdot (2^x) + 2\sqrt{2} = 0$$

$$\begin{cases} 2^\alpha + 2^\beta = 10 \dots\dots\dots (1) \\ 2^\alpha \cdot 2^\beta = 2\sqrt{2} = 2^{\frac{3}{2}} \dots (2) \end{cases}$$

由②得 $\alpha + \beta = \frac{3}{2}$

5. $\begin{cases} y = \left(\frac{1}{2}\right)^{|x|} \\ y = \frac{1}{5} \end{cases}$ 有兩個交點



$\therefore \left(\frac{1}{2}\right)^{|x|} = \frac{1}{5}$ 有 2 個實根

6. 設 $P(\alpha, 3)$ 、 $Q(\beta, 6)$

可得 $4^\alpha = 3 \dots\dots$ ①, $4^\beta = 6 \dots\dots$ ②

且直線 PQ 的斜率為 $\frac{6-3}{\beta-\alpha}$

① 得 $\frac{4^\beta}{4^\alpha} = 4^{\beta-\alpha} = \frac{6}{3} = 2 = 4^{\frac{1}{2}}$, 即 $\beta - \alpha = \frac{1}{2}$

故直線 PQ 的斜率為 $\frac{6-3}{\beta-\alpha} = \frac{3}{\frac{1}{2}} = 6$

7. (1) 將不等式化為 $2^{x^2} < 2^4$, 因為底數 $2 > 1$, 且以 2 為底的指數函數為嚴格遞增函數, 所以 $x^2 < 4$, 解得 $-2 < x < 2$

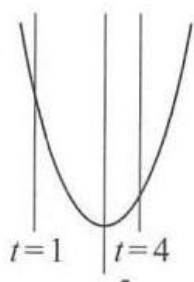
(2) 令 $t = \left(\frac{1}{3}\right)^x$, 不等式可化為 $4t - t^2 - 3 < 0$,

即 $(t-3)(t-1) > 0$

解得 $t > 3$ 或 $t < 1$

即 $\left(\frac{1}{3}\right)^x > 3 = \left(\frac{1}{3}\right)^{-1}$ 或 $\left(\frac{1}{3}\right)^x < 1 = \left(\frac{1}{3}\right)^0$

$\therefore$ 底數 $\frac{1}{3} < 1 \therefore x < -1$ 或 $x > 0$



$$t = 3$$

8. 令 $t = 2^x (t > 0)$

$\therefore 0 \leq x \leq 2$ 且以 2 為底的指數函數為嚴格遞增函數

$\therefore 2^0 \leq 2^x \leq 2^2$, 即 $1 \leq t \leq 4$

將 $f(x) = 4^x - 6 \times 2^x$ 改寫為

$$t^2 - 6t = (t - 3)^2 - 9 \quad , \text{ 其中 } 1 \leq t \leq 4$$

故當 $t = 1$ 時, $f(x)$ 有最大值-5 ; 當 $t = 3$ 時, $f(x)$ 有最小值-9

9. 令 $t = 3^x + 3^{-x}$, 則 $t \geq 2$

(由算幾不等式, $\frac{3^x + 3^{-x}}{2} \geq \sqrt{3^x \times 3^{-x}}$)

$$\Rightarrow \text{原函數可改寫成 } g(t) = (t^2 - 2) - 2t + 7 = t^2 - 2t + 5$$

$$= (t - 1)^2 + 4 \quad (\text{其中 } t \geq 2)$$

$\therefore$ 當 $t = 2$ 時, $g(2) = 5$ 為最小值, 此時 $3^x = 3^{-x} = 1$, 故 $x = 0$

10. $(x + 1) - p = x + 3, (-1) + q = 2 \Rightarrow p = -2, q = 3 \therefore$ 數對 $(p, q) = (-2, 3)$